

LECTURE NO. 19

- Relative Frequency Definition of Probability
- Axiomatic Definition of Probability
- Laws of Probability
 - Rule of Complementation
 - Addition Theorem

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THE RELATIVE FREQUENCY DEFINITION OF PROBABILITY ('A POSTERIORI' DEFINITION OF PROBABILITY)

based on past experience and experiments

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If a random experiment is repeated a large number of times, say n times, under identical conditions and if an event A is observed to occur m times, then the probability of the event A is defined as the LIMIT of the relative frequency m/n as n tends to infinity.

Symbolically, we write

$$P(A) = \lim_{n \rightarrow \infty} \frac{m}{n}$$

The definition assumes that as n increases indefinitely, the ratio m/n tends to become stable at the numerical value $P(A)$. The relationship between relative frequency and probability can also be represented as follows:

Relative Frequency $\rightarrow$ Probability mcq

as $n \rightarrow \infty$

As its name suggests, the relative frequency definition relates to the relative frequency with which an event occurs in the long run. In situations where we can say that an experiment has been repeated a very large number of times, the relative frequency definition can be applied.

As such, this definition is very useful in those practical situations where we are interested in computing a probability in numerical form but where the classical definition cannot be applied. (Numerous real-life situations are such where various possible outcomes of an experiment are NOT equally likely). This type of probability is also called empirical probability as it is based on EMPIRICAL evidence i.e. on OBSERVATIONAL data.

It can also be called **STATISTICAL PROBABILITY** for it is this very probability that forms the basis of mathematical statistics. mcq

Let us try to understand this concept by means of two examples:

- from a coin-tossing experiment and
- From data on the numbers of boys and girls born.

EXAMPLE-1

$$5067/10,000 = 0.5067$$

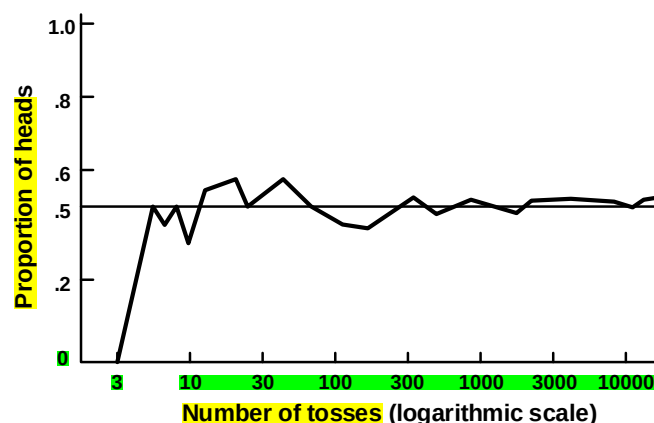
$$4933/10,000 = 0.4933$$

Coin-Tossing:

No one can tell which way a coin will fall but we expect the proportion of heads and tails after a large no. of tosses to be nearly equal. An experiment to demonstrate this point was performed by **Kerrich in Denmark in 1946**. He tossed a coin **10,000 times**, and obtained altogether **5067 heads** and **4933 tails**.

The behavior of the proportion of heads throughout the experiment is shown as in the following figure:

The proportion; of heads in a sequence of tosses of a coin (Kerrich, 1946):



As you can see, the curve fluctuates *widely* at first, but begins to settle down to a more or less *stable* value as the number of spins increases. It seems reasonable to suppose that the fluctuations would continue to diminish if the experiment were continued *indefinitely*, and the proportion of heads would cluster more and more *closely* about a *limiting* value which would be *very near*, if not exactly, one-half. This hypothetical *limiting* value is the (statistical) probability of heads.

Let us now take an example closely related to our daily lives --- that relating to the sex ratio:-

In this context, the first point to note is that it has been known since the eighteenth century that in *reliable* birth statistics based on sufficiently *large* numbers (in at least some parts of the world), there is always a slight *excess* of boys. Laplace records that, among the 215,599 births in thirty districts of France in the years 1800 to 1802, there were 110,312 boys and 105,287 girls. The proportions of boys and girls were thus 0.512 and 0.488 respectively (indicating a slight excess of boys over girls). In a *smaller* number of births one would, however, expect considerable *deviations* from these proportions. This point can be illustrated with the help of the following example:

EXAMPLE-2

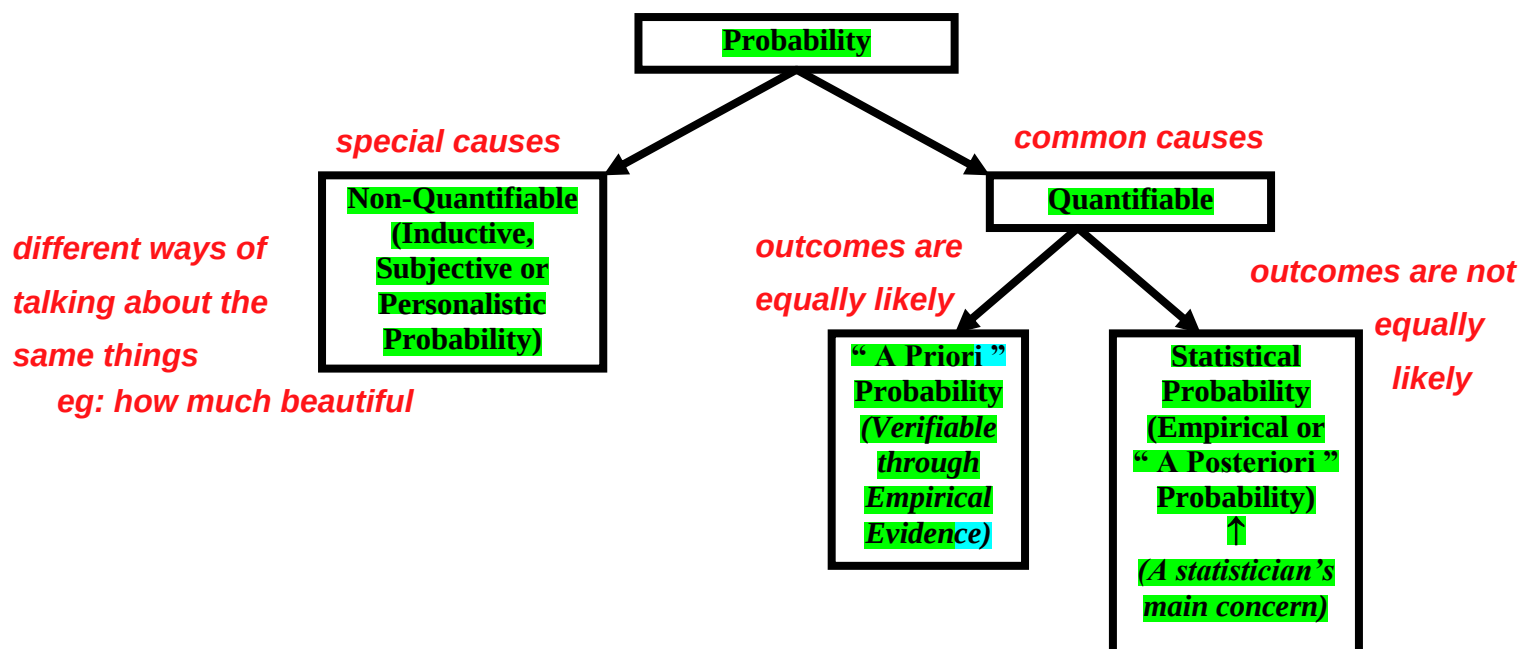
The following table shows the proportions of male births that have been worked out for the major regions of England as well as the rural districts of Dorset (for the year 1956).

Proportions of Male Births in various Regions and Rural Districts of England in 1956

Region of England	Proportion of Male Births	Rural Districts of Dorset	Proportion of Male Births
Northern	.514	Beaminster	.38
E. & W. Riding	.513	Blandford	.47
North Western	.512	Bridport	.53
North Midland	.517	Dorchester	.50
Midland	.514	Shaftesbury	.59
Eastern	.516	Sherborne	.44
London and S. Eastern	.514	Sturminster	.54
Southern	.514	Wareham and Purbeck	.53
South Western	.513	Wimborne & Cranborne	.54
Whole country	.514	All Rural District's of Dorset	.512

(Source: Annual Statistical Review)

As you can see, the figures for the rural districts of Dorset, based on about 200 births each, fluctuate between 0.38 and 0.59. While those for the major regions of England, which are each based on about 100,000 births, do not fluctuate much, rather, they range between 0.512 and 0.517 only. The larger sample size is clearly the reason for the greater constancy of the latter. We can imagine that if the sample were increased indefinitely, the proportion of boys would tend to a *limiting value* which is unlikely to differ much from 0.514, the proportion of male births for the *whole* country. This hypothetical *limiting* value is the (statistical) *probability* of a male birth. The overall discussion regarding the various ways in which probability can be defined is presented in the following diagram:



As far as *quantifiable* probability is concerned, in those situations where the various possible outcomes of our experiment are equally likely, we can compute the probability *prior* to actually conducting the experiment --- otherwise, as is generally the case, we can compute a probability only *after* the experiment has been conducted (and this is why it is also called ‘a posteriori’ probability). *Non-quantifiable probability is the one that is called Inductive Probability. It refers to the degree of belief which it is reasonable to place in a proposition on given evidence.*

An important point to be noted is that it is difficult to express inductive probabilities numerically — to construct a numerical scale of inductive probabilities, with 0 standing for impossibility and for logical certainty. An important point to be noted is that it is difficult to express inductive probabilities numerically — to construct a numerical scale of inductive probabilities, with 0 standing for impossibility and for logical certainty. Most statisticians have arrived at the conclusion that inductive probability cannot, in general, be measured and, therefore cannot be used in the mathematical theory of statistics.

This conclusion is not, perhaps, very surprising since there seems no reason why rational degree of belief should be measurable any more than, say, degrees of beauty. Some paintings are very beautiful, some are quite beautiful, and some are ugly, but it would be observed to try to construct a numerical scale of beauty, on which Mona Lisa had a beauty value of 0.96. Similarly some propositions are highly probable, some are quite probable and some are improbable, but it does not seem possible to construct a numerical scale of such (inductive) probabilities. Because of the fact that inductive probabilities are not quantifiable and cannot be employed in a mathematical argument, this is the reason why the usual methods of statistical inference such as tests of significance and confidence interval are based entirely on the concept of statistical probability. Although we have discussed three different ways of defining probability, the most formal definition is yet to come.

This is The Axiomatic Definition of Probability.

THE AXIOMATIC DEFINITION OF PROBABILITY

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This definition, introduced in 1933 by the Russian mathematician Andrei N. Kolmogorov, is based on a set of AXIOMS. Let S be a sample space with the sample points $E_1, E_2, \dots, E_i, \dots, E_n$. To each sample point, we assign a real number, denoted by the symbol $P(E_i)$, and called the probability of E_i , that must satisfy the following basic axioms:

Axiom 1:

For any event E_i ,
 $0 < P(E_i) < 1$

Axiom 2:

$P(S) = 1$
for the sure event S .

Axiom 3:

If A and B are mutually exclusive events (subsets of S), then

events are disjoint

$$P(A \cup B) = P(A) + P(B)$$

It is to be emphasized that According to the axiomatic theory of probability:

SOME probability defined as a non-negative real number is to be ATTACHED to each sample point E_i such that the sum of all such numbers must equal ONE. The ASSIGNMENT of probabilities may be based on past evidence or on some other underlying conditions. (If this assignment of probabilities is based on past evidence, we are talking about EMPIRICAL probability, and if this assignment is based on underlying conditions that ensure that the various possible outcomes of a random experiment are EQUALLY LIKELY, then we are talking about the CLASSICAL definition of probability. Let us consider another example:

EXAMPLE

Table given below shows the numbers of births in England and Wales in 1956 classified by (a) sex and (b) whether live born or stillborn.

Table-1

Number of births in England and Wales in 1956 by sex and whether live- or still born
(Source Annual Statistical Review)

	Liveborn	Stillborn	Total
Male	359,881 (A)	8,609 (B)	368,490
Female	340,454 (C)	7,796 (D)	348,250
Total	700,335	16,405	716,740

There are four possible events in this double classification:

- Male livebirth (denoted by A),
- Male stillbirth (denoted by B),
- Female livebirth (denoted by C)
- Female stillbirth (denoted by D),

The relative frequencies corresponding to the figures of Table-1 are given in Table-2:

Table-2

Proportion of births in England and Wales in 1956 by sex and whether live- or stillborn
(Source Annual Statistical Review)

	Liveborn	Stillborn	Total
Male	.5021	.0120	.5141
Female	.4750	.0109	.4859
Total	.9771	.0229	1.0000

The total number of births is large enough for these relative frequencies to be treated for all practical purposes as PROBABILITIES.

Let us denote the compound events 'Male birth' and 'Stillbirth' by the letters M and S. Now a male birth occurs whenever either a male live birth or a male stillbirth occurs, and so the proportion of male birth, regardless of whether they are live-or stillborn, is equal to the sum of the proportions of these two types of birth; that is to say,

$$p(M) = p(A \text{ or } B) = p(A) + p(B) \\ = .5021 + .0120 = .5141$$

Similarly, a stillbirth occurs whenever either a male stillbirth or a female stillbirth occurs and so the proportion of stillbirths, regardless of sex, is equal to the sum of the proportions of these two events:

$$p(S) = p(B \text{ or } D) = p(B) + p(D) \\ = .0120 + .0109 = .0229$$

Let us now consider some basic LAWS of probability. These laws have important applications in solving probability problems.

LAW OF COMPLEMENTATION

If $\bar{A}$ is the complement of an event A relative to the sample space S, then

$$P(\bar{A}) = 1 - P(A)$$

Hence the probability of the complement of an event is equal to one minus the probability of the event. Complementary probabilities are very useful when we are wanting to solve questions of the type ‘What is the probability that, in tossing two fair dice, at least one even number will appear?’

EXAMPLE

A coin is tossed 4 times in succession. What is the probability that at least one head occurs?

- The sample space S for this experiment consists of $2^4 = 16$ sample points (as each toss can result in 2 outcomes), and
- We assume that each outcome is equally likely.

If we let A represent the event that at least one head occurs, then A will consist of MANY sample points, and the process of computing the probability of this event will become somewhat cumbersome! So, instead of denoting this particular event by A, let us denote its complement i.e. “No head” by $\bar{A}$.

Thus the event $\bar{A}$ consists of the SINGLE sample point {TTTT}.

Therefore $P(\bar{A}) = 1/16$.

Hence by the law of complementation, we have

$$P(A) = 1 - P(\bar{A}) = 1 - \frac{1}{16} = \frac{15}{16}$$

The next law that we will consider is the Addition Law or the General Addition Theorem of Probability:

ADDITION LAW

If A and B are any two events defined in a sample space S, then

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

In words, this law may be stated as follows:

“If two events A and B are not mutually exclusive, then the probability that at least one of them occurs, is given by the sum of the separate probabilities of events A and B minus the probability of the joint event $A \cap B$.”

LECTURE NO. 20

- Application of Addition Theorem
- Conditional Probability
- Multiplication Theorem

First of all, let us consider in some detail the Addition Law or the General Addition Theorem of Probability:

ADDITION LAW

If A and B are any two events defined in a sample space S, then $P(A \cup B) = P(A) + P(B) - P(A \cap B)$

In words, this law may be stated as follows:

“If two events A and B are not mutually exclusive, then the probability that at least one of them occurs, is given by the sum of the separate probabilities of events A and B minus the probability of the joint event $A \cap B$.”

EXAMPLE

If one card is selected at random from a deck of 52 playing cards, what is the probability that the card is a club or a face card or both?

Let A represent the event that the card selected is a club, B, the event that the card selected is a face card, and $A \cap B$, the event that the card selected is both a club and a face card. Then we need $P(A \cup B)$

Now $P(A) = 13/52$, as there are 13 clubs, $P(B) = 12/52$, as there are 12 face cards,

$P(A \cap B) = 3/52$, since 3 of clubs are also face cards.

Therefore the desired probability is

$$\begin{aligned} P(A \cup B) &= P(A) + P(B) - P(A \cap B) \\ &= 13/52 + 12/52 - 3/52 \\ &= 22/52. \end{aligned}$$

COROLLARY-1

If A and B are mutually exclusive events, then

$$P(A \cup B) = P(A) + P(B)$$

(Since $A \cap B$ is an impossible event, hence $P(A \cap B) = 0$)

EXAMPLE

Suppose that we toss a pair of dice, and we are interested in the event that we get a total of 5 or a total of 11. What is the probability of this event?

SOLUTION

if they can not occur at the same time.

In this context, the first thing to note is that ‘getting a total of 5’ and ‘getting a total of 11’ are mutually exclusive events. Hence, we should apply the special case of the addition theorem. If we denote ‘getting a total of 5’ by A, and ‘getting a total of 11’ by B, then $P(A) = 4/36$ (since there are four outcomes favorable to the occurrence of a total of 5), and $P(B) = 2/36$ (since there are two outcomes favorable to the occurrence of a total of 11).

Hence the probability that we get a total of 5 or a total of 11 is given by $(1,2,3,4,5,6) (1,2,3,4,5,6)$

$$P(A \cup B) = P(A) + P(B)$$

$$= 4/36 + 2/36 = 6/36 = 16.67\%.$$

(1, 1), (1, 2), (1, 3), (1, 4), (1, 5), (1, 6), (2, 1), (2, 2), (2, 3), (2, 4), (2, 5), (2, 6), (3, 1), (3, 2), (3, 3), (3, 4), (3, 5), (3, 6), (4, 1), (4, 2), (4, 3), (4, 4), (4, 5), (4, 6), (5, 1), (5, 2), (5, 3), (5, 4), (5, 5), (5, 6), (6, 1), (6, 2), (6, 3), (6, 4), (6, 5), (6, 6)

COROLLARY-2

If $A_1, A_2, \dots, A_k$ are k mutually exclusive events, then the probability that one of them occurs, is the sum of the probabilities of the separate events, i.e.

$$P(A_1 \cup A_2 \cup \dots \cup A_k) = P(A_1) + P(A_2) + \dots + P(A_k).$$

Let us now consider an interesting example to illustrate the way in which probability problems can be solved:

EXAMPLE

Three horses A, B and C are in a race; A is twice as likely to win as B and B is twice as likely to win as C. What is the probability that A or B wins?

Evidently, the events mentioned in this problem are not equally likely.

Let $P(C) = p$

Then $P(B) = 2p$ as B is twice as likely to win as C.

Similarly

$$P(A) = 2P(B) = 2(2p) = 4p$$

In this problem, we assume that no two of the horses A, B and C cannot win the race together (i.e. the race cannot end in a draw).

Hence, the events A, B and C are mutually exclusive.

Since A, B and C are mutually exclusive and collectively *exhaustive*, therefore the sum of their probabilities must be equal to 1.

Thus

$$p + 2p + 4p = 1$$

$$\text{or } p = 1/7$$

$$\therefore P(C) = 1/7,$$

$$P(B) = 2(1/7) = 2/7,$$

$$\text{and } P(A) = 4(1/7) = 4/7.$$

Hence

$$\begin{aligned} P(A \cup B) &= P(A) + P(B) \\ &= 4/7 + 2/7 \\ &= 6/7. \end{aligned}$$

Having discussed the addition theorem in some detail, we would now like to discuss the Multiplication Theorem.

But, before we are in a position to take up the multiplication theorem, we need to consider the concept of conditional probability.

CONDITIONAL PROBABILITY

The sample space for an experiment must often be changed when some additional information pertaining to the outcome of the experiment is received.

The effect of such information is to REDUCE the sample space by excluding some outcomes as being impossible which BEFORE receiving the information were believed possible. The probabilities associated with such a reduced sample space are called conditional probabilities.

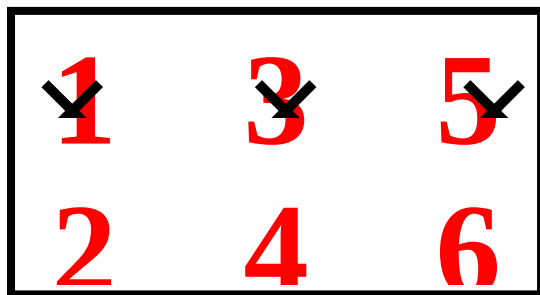
The following example illustrates the concept of conditional probability

EXAMPLE

Suppose that we toss a fair die. Then the sample space of this experiment is $S = \{1, 2, 3, 4, 5, 6\}$. Suppose we wish to know the probability of the outcome that the die shows 6 (say event A). Also, suppose that, before seeing the outcome, we are told that the die shows an EVEN number of dots (say event B).

Then the information that the die shows an even number excludes the outcomes 1, 3 and 5, and thereby reduces the original sample space to a sample space that consists of three outcomes 2, 4 and 6, i.e. the reduced sample space is

$$B = \{2, 4, 6\}.$$



(The sample space is reduced.)

Then, the desired probability in the reduced sample space B is $1/3$.

(since each outcome in the reduced sample space is EQUALLY LIKELY). This probability $1/3$ is called the conditional probability of the event A because it is computed under the CONDITION that the die has shown an even number of dots. In other words,

$$P(\text{die shows 6} | \text{die shows even numbers})$$

$$= 1/3,$$

(Where the vertical line is read as given that and the information following the vertical line describes the conditioning event).

Sometimes, it is not very convenient to compute a conditional probability by first determining the number of sample points that belong to the reduced sample space.

In such a situation, we can utilize the following *alternative method of computing a conditional probability*

the probability of an event occurring given that another event has already occurred

CONDITIONAL PROBABILITY

If A and B are two events in a sample space S and if P(B) is not equal to zero, then the conditional probability of the event A given that event B has occurred, written as P(A/B), is defined by

$$P(A/B) = \frac{P(A \cap B)}{P(B)}$$

Where $P(B) > 0$

(If $P(B) = 0$, the conditional probability P(A/B) remains undefined.)

Similarly

$$P(B/A) = \frac{P(A \cap B)}{P(A)}$$

where $P(A) > 0$.

It should be noted that P(A/B) SATISFIES all the basic axioms of probability, namely:

- $0 < P(A/B) < 1$.
- ii) $P(S/B) = 1$
- $P(A_1 \cup A_2/B) = P(A_1/B) + P(A_2/B)$ (provided that the events A_1 and A_2 are mutually exclusive).

Let us now apply this concept to a real-world example

EXAMPLE-2

At a certain elementary school in a Western country, the school-record of the past ten years shows that 75% of the students come from a two-parent home and that 20% of the students are low-achievers and belong to two-parent homes. What is the probability that such a randomly selected student will be a low achiever GIVEN THAT he or she comes from a two-parent home?

SOLUTION

Let A denote a low achiever and B a student from a two-parent home. Applying the relative frequency definition of probability, we have

$$P(B) = 0.75 \text{ and } P(A \cap B) = 0.20.$$

Thus, we obtain

$$P(A/B) = \frac{P(A \cap B)}{P(B)} = \frac{0.20}{0.75} = 0.27$$

MULTIPLICATION THEOREM OF PROBABILITY

It is interesting to note that the multiplication theorem is obtained very conveniently from the formula of conditional probability:

$$P(A/B) = \frac{P(A \cap B)}{P(B)}$$

As discussed earlier, the conditional probability of A given that B has occurred has already been defined as:

$$P(A/B) = \frac{P(A \cap B)}{P(B)}, \text{ Where } P(B) > 0$$

Multiplying both sides by P(B), we get

$$P(A \cap B) = P(B) \cdot P(A/B).$$

And if we interchange the roles of A and B, we obtain:

$$P(A \cap B) = P(A) P(B/A),$$

Provided $P(A) > 0$.

MULTIPLICATION LAW

If A and B are any two events defined in a sample space S, then

$$P(A \cap B)$$

$$= P(A) P(B/A), \text{ provided } P(A) > 0,$$

$$= P(B) P(A/B) \text{ provided } P(B) > 0.$$

(The second form is easily obtained by interchanging A and B. This is called the GENERAL rule of multiplication of probabilities. It can be stated as follows:

MULTIPLICATION LAW

“The probability that two events A and B will both occur is equal to the probability that one of the events will occur multiplied by the conditional probability that the other event will occur given that the first event has already occurred.”
Let us apply the concept of multiplication theorem to an example

EXAMPLE

A box contains 15 items, 4 of which are defective and 11 is good. Two items are selected. What is the probability that the first is good and the second defective?

Let A represent the event that the first item selected is good, and B, the event that the second items is defective.

Then we need to calculate the probability of the JOINT event $A \cap B$ by the rule

$$P(A \cap B) = P(A)P(B/A).$$

We have:

Type of Item	No. of Items
Defective	4
Good	11
Total	15

Since all the items are equally likely to be chosen, hence

$$P(A) = 11/15.$$

Given the event A has occurred, there remain 14 items of which 4 are defective. Therefore the probability of selecting a defective item after a good item has been selected is 4/14 i.e.

$$P(B/A) = 4/14.$$

Hence

$$\begin{aligned} P(A \cap B) &= P(A) P(B/A) \\ &= 11/15 \times 4/14 \\ &= 44/210 \\ &= 0.16. \end{aligned}$$

In this lecture, the concepts of the Addition Theorem and the Multiplication Theorem of probability have been discussed in some detail.

In order to **differentiate** between the situation where the addition theorem is applicable and the situation where the multiplication theorem is applicable, the main point to keep in mind is that whenever we wish to compute the probability that *either* A occurs *or* B occurs, we should think of the Addition Theorem, where as, whenever we wish to compute the probability that *both* A and B occur, we should think of the Multiplication Theorem.

LECTURE NO. 21

- Independent and Dependent Events
- Multiplication Theorem of Probability for Independent Events
- Marginal Probability

Before we proceed the concept of independent versus dependent events, let us review the Addition and Multiplication Theorems of Probability that were discussed in the last lecture.

To this end, let us consider an interesting example that illustrates the application of both of these theorems in one problem:

EXAMPLE

A bag contains 10 white and 3 black balls. Another bag contains 3 white and 5 black balls. Two balls are transferred from first bag and placed in the second, and then one ball is taken from the latter.

What is the probability that it is a white ball?

In the beginning of the experiment, we have:

Colour of Ball	No. of Balls in Bag A	No. of Balls in Bag B
White	10	3
Black	3	5
Total	13	8

Let A represent the event that 2 balls are drawn from the first bag and transferred to the second bag. Then A can occur in the following three mutually exclusive ways:

A₁ = 2 white balls are transferred to the second bag.

A₂ = 1 white ball and 1 black ball are transferred to the second bag.

A₃ = 2 black balls are transferred to the second bag.

Then, the total number of ways in which 2 balls can be drawn out of a total of 13 balls is $\binom{13}{2}$.

And, the total number of ways in which 2 white balls can be drawn out of 10 white balls is $\binom{10}{2}$.

Thus, the probability that two white balls are selected from the first bag containing 13 balls (in order to transfer to the second bag) is

$$P(A_1) = \frac{\binom{10}{2}}{\binom{13}{2}} = \frac{45}{78},$$

Similarly, the probability that one white ball and one black ball are selected from the first bag containing 13 balls (in order to transfer to the second bag) is

$$P(A_2) = \frac{\binom{10}{1} \binom{3}{1}}{\binom{13}{2}} = \frac{30}{78},$$

And, the probability that two black balls are selected from the first bag containing 13 balls (in order to transfer to the second bag) is

$$P(A_3) = \frac{\binom{3}{2}}{\binom{13}{2}} = \frac{3}{78}.$$

AFTER having transferred 2 balls from the first bag, the second bag contains

i) 5 white and 5 black balls (if 2 white balls are transferred)

Colour of Ball	No. of Balls in Bag A	No. of Balls in Bag B
White	10 – 2 = 8	3 + 2 = 5
Black	3	5
Total	13 – 2 = 11	8 + 2 = 10

Hence: $P(W/A1) = 5/10$

ii) 4 white and 6 black balls (if 1 white and 1 black ball are transferred)

Colour of Ball	No. of Balls in Bag A	No. of Balls in Bag B
White	$10 - 1 = 9$	$3 + 1 = 4$
Black	$3 - 1 = 2$	$5 + 1 = 6$
Total	$13 - 2 = 11$	$8 + 2 = 10$

Hence: $P(W/A2) = 6/10$

iii) 3 white and 7 black balls (if 2 black balls are transferred)

Colour of Ball	No. of Balls in Bag A	No. of Balls in Bag B
White	10	3
Black	$3 - 2 = 1$	$5 + 2 = 7$
Total	$13 - 2 = 11$	$8 + 2 = 10$

Hence: $P(W/A3) = 3/10$

Let W represent the event that the WHITE ball is drawn from the second bag after having transferred 2 balls from the first bag.

Then $P(W) = P(A1 \cap W) + P(A2 \cap W) + P(A3 \cap W)$

Now $P(A1 \cap W) = P(A1) P(W/A1)$

$$= 45/78 \times 5/10$$

$$= 15/52$$

$P(A2 \cap W) = P(A2) P(W/A2)$

$$= 30/78 \times 4/10$$

$$= 2/13,$$

And

$P(A3 \cap W) = P(A3) P(W/A3)$

$$= 3/78 \times 3/10$$

$$= 3/260.$$

Hence the required probability is

$P(W)$

$$= P(A1 \cap W) + P(A2 \cap W) + P(A3 \cap W)$$

$$= 15/52 + 2/13 + 3/260$$

$$= 59/130$$

$$= 0.45$$

INDEPENDENT EVENTS

Two events A and B in the same sample space S, are defined to be independent (or statistically independent) if the probability that one event occurs, is not affected by whether the other event has or has not occurred, that is

$P(A/B) = P(A)$ and $P(B/A) = P(B)$. It then follows that two events A and B are independent if and only if

$$P(A \cap B) = P(A) P(B)$$

and this is known as the special case of the Multiplication Theorem of Probability.

RATIONALE

According to the multiplication theorem of probability, we have:

$$P(A \cap B) = P(A) \cdot P(B/A)$$

Putting $P(B/A) = P(B)$, we obtain

$$P(A \cap B) = P(A) P(B)$$

The events A and B are defined to be DEPENDENT if $P(A \cap B) \neq P(A) \times P(B)$. MCQ

This means that the occurrence of one of the events in some way affects the probability of the occurrence of the other event. Speaking of independent events, it is to be emphasized that two events that are independent, can NEVER be mutually exclusive. MCQ

EXAMPLE

Two fair dice, one red and one green, are thrown. Let A denote the event that the red die shows an even number and let B denote the event that the green die shows a 5 or a 6. Show that the events A and B are independent.

The sample space S is represented by the following 36 outcomes:

S = {(1, 1), (1, 2), (1, 3), (1, 5), (1, 6);
 (2, 1), (2, 2), (2, 3), (2, 5), (2, 6);
 (3, 1), (3, 2), (3, 3), (3, 5), (3, 6);
 (4, 1), (4, 2), (4, 3), (4, 5), (4, 6);
 (5, 1), (5, 2), (5, 3), (5, 5), (5, 6);
 (6, 1), (6, 2), (6, 3), (6, 5), (6, 6)}.

Since

A represents the event that red die shows an even number, and B represents the event that green die shows a 5 or a 6, Therefore $A \cap B$ represents the event that red die shows an even number and green die shows a 5 or a 6.

Since A represents the event that red die shows an even number, hence $P(A) = 3/6$. Similarly, since B represents the event that green die shows a 5 or a 6, hence $P(B) = 2/6$.

Now, in order to compute the probability of the joint event $A \cap B$, the first point to note is that, in all, there are 36 possible outcomes when we throw the two dice together, i.e.

S = (1, 1), (1, 2), (1, 3), (1, 5), (1, 6);
 (2, 1), (2, 2), (2, 3), (2, 5), (2, 6);
 (3, 1), (3, 2), (3, 3), (3, 5), (3, 6);
 (4, 1), (4, 2), (4, 3), (4, 5), (4, 6);
 (5, 1), (5, 2), (5, 3), (5, 5), (5, 6);
 (6, 1), (6, 2), (6, 3), (6, 5), (6, 6)

Event A represents the red die showing an even number, which consists of the outcomes:

(2, 1), (2, 2), (2, 3), (2, 4), (2, 5), (2, 6)
 (4, 1), (4, 2), (4, 3), (4, 4), (4, 5), (4, 6)
 (6, 1), (6, 2), (6, 3), (6, 4), (6, 5), (6, 6)

The joint event $A \cap B$ contains only 6 outcomes out of the 36 possible outcomes. There are a total of 18 outcomes in event A. $P(A) = 18/36 = 1/2$

These are (2, 5), (4, 5), (6, 5), (2, 6), (4, 6), and (6, 6).

and $P(A \cap B) = 6/36 = 1/6$

Event B represents the green die showing a 5 or a 6, which consists of the outcomes:

(1, 5), (1, 6), (2, 5), (2, 6), (3, 5), (3, 6), (4, 5), (4, 6), (5, 5), (5, 6), (6, 5), (6, 6)

Now

$P(A) P(B)$

$$= 3/6 \times 2/6 = 1/2 \times 1/3 = 1/6$$

$$= 6/36$$

$$= P(A \cap B).$$

There are a total of 12 outcomes in event B. $P(B) = 12/36 = 1/3$

Therefore the events A and B are independent. Let us now go back to the example pertaining to live births and stillbirths that we considered in the last lecture, and try to determine whether or not sex of the baby and nature of birth are independent.

EXAMPLE

Table-1 below shows the numbers of births in England and Wales in 1956 classified by (a) sex and (b) whether live born or stillborn.

Table-1

Number of births in England and Wales in 1956 by sex and whether live- or still born
 (Source Annual Statistical Review)

	Liveborn	Stillborn	Total
Male	359,881 (A)	8,609 (B)	368,490
Female	340,454 (B)	7,796 (D)	348,250
Total	700,335	16,405	716,740

There are four possible events in this double classification:

- Male live birth,
- Male stillbirth,
- Female live birth, and
- Female stillbirth.

The corresponding relative frequencies are given in Table-2.

Table-2

Proportion of births in England and Wales in 1956 by sex and whether live- or stillborn
 (Source Annual Statistical Review)

	Liveborn	Stillborn	Total
Male	.5021	.0120	.5141
Female	.4750	.0109	.4859
Total	.9771	.0229	1.0000

As discussed in the last lecture, the total number of births is large enough for these relative frequencies to be treated for all practical purposes as *PROBABILITIES*. The compound events ‘Male birth’ and ‘Stillbirth’ may be represented by the letters M and S. If M represents a male birth and S a stillbirth, we find that

$$\frac{n(M \text{ and } S)}{n(M)} = \frac{8609}{368490} = 0.0234$$

This figure is the proportion — and, since the sample size is large, it can be regarded as the *probability* — of males who are still born — in other words, the *CONDITIONAL* probability of a stillbirth *given that* it is a male birth. In other words, the probability of *stillbirths in males*. The corresponding proportion of stillbirths among females is

$$\frac{7796}{348258} = 0.0224.$$

These figures should be contrasted with the OVERALL, or *UNCONDITIONAL*, proportion of stillbirths, which is

$$\frac{16405}{716740} = 0.0229.$$

We observe that the conditional probability of stillbirths among boys is slightly *HIGHER* than the overall proportion. Where as the conditional proportion of stillbirths among girls is slightly *LOWER* than the overall proportion. It can be concluded that sex and stillbirth are statistically *DEPENDENT*, that is to say, the *SEX of a baby yet to be born has an effect, (although a small effect), on* its chance of being stillborn. The example, that we just considered point out the concept of *MARGINAL PROBABILITY*.

Let us have another look at the data regarding the live births and stillbirths in England and Wales:

Table-2 Proportion of births in England and Wales in 1956 by sex and whether live- or stillborn (Source *Annual Statistical Review*)

	Liveborn	Stillborn	Total
Male	.5021	.0120	.5141
Female	.4750	.0109	.4859
Total	.9771	.0229	1.0000

And, the figures in Table-2 indicate that the probability of male birth is 0.5141, whereas the probability of female birth is 0.4859. Also, the probability of live birth is 0.9771, where as the probability of stillbirth is 0.0229. And since these probabilities appear in the margins of the Table, they are known as *Marginal Probabilities*. According to the above table, the probability that a new born baby is a male and is live born is 0.5021 whereas the probability that a new born baby is a male and is stillborn is 0.0120. Also, as stated earlier; the probability that a new born baby is a male is 0.5141, and, CLEARLY, $0.5141 = 0.5021 + 0.0120$. Hence, it is clear that the joint probabilities occurring in any row of the table *ADD UP* to yield the corresponding *marginal* probability. If we reflect upon this situation carefully, we will realize that this equation is totally in accordance with the Addition Theorem of Probability for mutually exclusive events.

$$\begin{aligned} P(\text{male birth}) &= P(\text{male live-born or male stillborn}) \\ &= P(\text{male live-born}) + P(\text{male stillborn}) \\ &= 0.5021 + 0.0120 \\ &= 0.5141 \end{aligned}$$

EXAMPLE

$$\begin{aligned} P(\text{stillbirth/male birth}) &= P(\text{male birth and stillbirth})/P(\text{male birth}) \\ &= 0.0120/0.5141 \\ &= 0.0233 \end{aligned}$$

LECTURE NO. 22

- Bayes' Theorem
- Discrete Random Variable
 - Discrete Probability Distribution
 - Graphical Representation of a Discrete Probability Distribution
 - Mean, Standard Deviation and Coefficient of Variation of a Discrete Probability Distribution
 - Distribution Function of a Discrete Random Variable.

First of all, let us discuss the BAYES' THEOREM. This theorem deals with conditional probabilities in an interesting way:

BAYES' THEOREM

it is used to calculate the conditional probability

If events $A_1, A_2, \dots, A_k$ form a PARTITION of a sample space S (that is, the events A_i are mutually exclusive and exhaustive (i.e. their union is S)), and if B is any other event of S such that it can occur ONLY IF ONE OF THE A_i OCCURS, then for any i ,

$$P(A_i | B) = \frac{P(A_i)P(B | A_i)}{\sum_{i=1}^k P(A_i)P(B | A_i)},$$

for $i = 1, 2, \dots, k$.

Stated differently:

BAYES' THEOREM:

If $A_1, A_2, \dots$ and A_k are mutually exclusive events of which one must occur, then

$$P(A_i | B) = \frac{P(A_i) \cdot P(B | A_i)}{P(A_1) \cdot P(B | A_1) + P(A_2) \cdot P(B | A_2) + \dots + P(A_k) \cdot P(B | A_k)}$$

If $k = 2$, we obtain:

Bayes' Theorem for two mutually exclusive events A_1 and A_2 :

$$P(A_i | B) = \frac{P(A_i) \cdot P(B | A_i)}{P(A_1) \cdot P(B | A_1) + P(A_2) \cdot P(B | A_2)}$$

Where $i = 1, 2$.

In other words

$$\begin{aligned} \text{and } P(A_1 | B) &= \frac{P(A_1) \cdot P(B | A_1)}{P(A_1) \cdot P(B | A_1) + P(A_2) \cdot P(B | A_2)} \\ P(A_2 | B) &= \frac{P(A_2) \cdot P(B | A_2)}{P(A_1) \cdot P(B | A_1) + P(A_2) \cdot P(B | A_2)} \end{aligned}$$

EXAMPLE

In a developed country where cars are tested for the emission of pollutants, 25 percent of all cars emit excessive amounts of pollutants. When tested, 99 percent of all cars that emit excessive amounts of pollutants will fail, but 17 percent of the cars that do not emit excessive amounts of pollutants will also fail. What is the probability that a car that fails the test actually emits excessive amounts of pollutants?

SOLUTION

Let A_1 denote the event that it emits EXCESSIVE amounts of pollutants, and let A_2 denote the event that a car does NOT emit excessive amounts of pollutants. (In other words, A_2 is the complement of A_1 .)

Also, let B denote the event that a car FAILS the test.

The first thing to note is that any car will either emit or not emit excessive amounts of pollutants. In other words, A_1 and A_2 are mutually exclusive and exhaustive events i.e. A_1 and A_2 form a PARTITION of the sample space S .

Hence, we are in a position to apply the Bayes' theorem.

We need to calculate $P(A_1|B)$, and, according to the Bayes' theorem:

$$P(A_1 | B) = \frac{P(A_1) \cdot P(B | A_1)}{P(A_1) \cdot P(B | A_1) + P(A_2) \cdot P(B | A_2)}$$

Now, according to the data given in this problem: $25\% = 25/100 = 0.25$

$$P(A_1) = 0.25,$$

$$P(A_2) = 0.75 \text{ (as } A_2 \text{ is simply the complement of } A_1),$$

$$P(B|A_1) = 0.99,$$

and

$$P(B|A_2) = 0.17$$

Substituting the above values in the Bayes' theorem, we obtain:

$$\begin{aligned} P(A_1 | B) &= \frac{P(A_1) \cdot P(B | A_1)}{P(A_1) \cdot P(B | A_1) + P(A_2) \cdot P(B | A_2)} \\ &= \frac{(0.25)(0.99)}{(0.25)(0.99) + (0.75)(0.17)} \\ &= \frac{0.2475}{0.2475 + 0.1275} \\ &= \frac{0.2475}{0.3750} \\ &= 0.66 \end{aligned}$$

This is the probability that a car which fails the test ACTUALLY emits excessive amounts of pollutants. The example that we just considered pertained to the simplest case when we have only two mutually exclusive and exhaustive events A_1 and A_2 .

As stated earlier, the Bayes' theorem can be extended to the case of three, four, five or more mutually exclusive and exhaustive events.

Let us consider another example: In the following example, check the percentages of defective bolts from the recorded lecture.

EXAMPLE

In a bolt factory, 25% of the bolts are produced by machine A, 35% are produced by machine B, and the remaining 40% are produced by machine C. Of their outputs, 2%, 4% and 5% respectively are defective bolts. If a bolt is selected at random and found to be defective, what is the probability that it came from machine A?

$$2\% = 0.02$$

In this example, we realize that "a bolt is produced by machine A", "a bolt is produced by machine B" and "a bolt is produced by machine C" represent three mutually exclusive and exhaustive events i.e. we can regard them as A_1 , A_2 and A_3 . The event "defective bolt" represents the event B. Hence, in this example, we need to determine $P(A_1|B)$.

The students are encouraged to work on this problem on their own, in order to understand the application and significance of the Bayes' Theorem. This brings us to the END of the discussion of various basic concepts of probability. We now begin the discussion of a very important concept in mathematical statistics, i.e., the concept of PROBABILITY DISTRIBUTIONS.

As stated in the very beginning of this course, there are two types of quantitative variables --- the discrete variable, and the continuous variable. Accordingly, we have the discrete probability distribution as well as the continuous probability distribution.

We begin with the discussion of the discrete probability distribution.

In this regard, the first concept that we need to consider is the concept of Random variable.

RANDOM VARIABLE

Such a numerical quantity whose value is determined by the outcome of a random experiment is called a random variable.

For example, if we toss three dice together, and let X denote the number of heads, then the random variable X consists of the values 0, 1, 2, and 3. Obviously, in this example, X is a discrete random variable. Let us now discuss the concept of discrete probability distribution in detail with the help of the following example:

Example:

If a biologist is interested in the number of petals on a particular flower, this number may take the values 3, 4, 5, 6, 7, 8, 9, and each one of these numbers will have its own probability.

Suppose that upon observing a large no. of flowers, say 1000 flowers, of that particular species, the following results are obtained:

No. of Petals X	f
3	50
4	100
5	200
6	300
7	250
8	75
9	25
	1000

Since 1000 is quite a large number, hence the proportions $f/\sum f$ can be regarded as probabilities and hence we can write

No. of Petals X	P(x)
$x_1 = 3$	0.05
$x_2 = 4$	0.10
$x_3 = 5$	0.20
$x_4 = 6$	0.30
$x_5 = 7$	0.25
$x_6 = 8$	0.075
$x_7 = 9$	0.025
	1

PROPERTIES OF A DISCRETE PROBABILITY DISTRIBUTION

- (1) $0 \leq P(X_i) \leq 1$ for each X_i ($i = 1, 2, \dots, 7$)
- (2) $\sum p(X_i) = 1$

And, since the number of petals on a leaf can only be a *whole* number, hence the variable X is known as a *discrete random variable*, and the probability distribution of this variable is known as a *DISCRETE probability distribution*. In other words, Any discrete variable that is associated with a random experiment, and attached to whose various values are various probabilities (Such that $\sum_{i=1}^{\infty} P(X_i) = 1$)

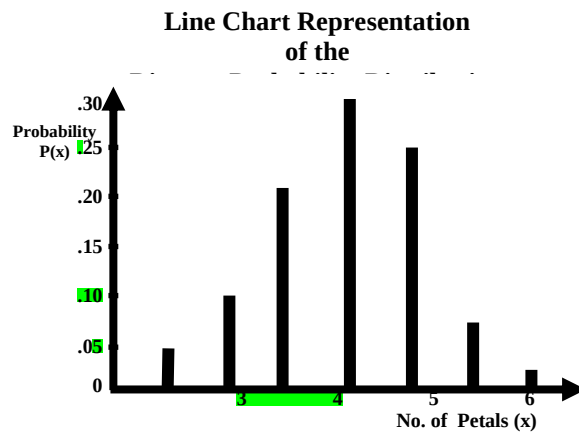
is known as a Discrete Random Variable, and its probability distribution is known as a Discrete Probability Distribution. Just as we can depict a frequency distribution graphically, we can draw the GRAPH of a probability distribution.

EXAMPLE

Going back to the probability distribution of the number of petals on the flowers of a particular species, i.e.:

No. of Petals X	P(x)
$x_1 = 3$	0.05
$x_2 = 4$	0.10
$x_3 = 5$	0.20
$x_4 = 6$	0.30
$x_5 = 7$	0.25
$x_6 = 8$	0.075
$x_7 = 9$	0.025
	1

This distribution can be represented in the form of a line chart.



Evidently, this particular probability distribution is approximately symmetric. In addition, this graph clearly shows that, just as in the case of a frequency distribution, every discrete probability distribution has a *CENTRAL* point and a *SPREAD*. Hence, similar to a frequency distribution, the discrete probability distribution has a *MEAN* and a *STANDARD DEVIATION*. How do we calculate the mean and the standard deviation of a probability distribution?

Let us first consider the computation of the MEAN:

We know that in the case of a frequency distribution such as

X	f
1	1
2	2
3	4
4	2
5	1

the mean is given by

$$\bar{X} = \frac{\sum fX}{\sum f} = \frac{\sum Xi}{\sum f}$$

In case of a discrete probability distribution, such as the one that we have been considering i.e

No. of Petals X	P(x)
$x_1 = 3$	0.05
$x_2 = 4$	0.10
$x_3 = 5$	0.20
$x_4 = 6$	0.30
$x_5 = 7$	0.25
$x_6 = 8$	0.075
$x_7 = 9$	0.025
	1

the mean is given by:

$$\mu = E(X) = \frac{\sum XP(X)}{\sum p(X)} = \frac{\sum XP(X)}{1} = \sum XP(X)$$

Hence we construct the column of $XP(X)$, as shown below:

No. of Petals x	P(x)	xP(x)
$x_1 = 3$	0.05	0.15
$x_2 = 4$	0.10	0.40
$x_3 = 5$	0.20	1.00
$x_4 = 6$	0.30	1.80
$x_5 = 7$	0.25	1.75
$x_6 = 8$	0.075	0.60
$x_7 = 9$	0.025	0.225
Total	1	5.925

Hence $\mu = E(X) = \sum XP(X) = 5.925$ i.e. the mean of the given probability distribution is 5.925. In other words, considering a very large number of flowers of that particular species, we would expect that, on the average, a flower contains 5.925 petals --- or, *rounding* this number, 6 petals. This interpretation points to the reason why the mean of the probability distribution of a random variable X is technically called the EXPECTED VALUE of the random variable X. ("Given that the probability that the flower has 3 petals is 5%, the probability that the flower has 4 petals is 10%, and so ON, we EXPECT that on the average a flower contains 5.925 petals.)

COMPUTATION OF THE STANDARD DEVIATION

Just as in case of a frequency distribution, we have

$$S = \sqrt{\frac{\sum f(X - \bar{X})^2}{\sum f}}$$

$$= \sqrt{\frac{\sum fX^2}{\sum f} - \left(\frac{\sum fX}{\sum f}\right)^2} = \sqrt{\frac{\sum X^2f}{\sum f} - \left(\frac{\sum Xf}{\sum f}\right)^2}$$

Similarly, in case of a probability distribution, we have

$$\sigma = \text{S.D.}(X) = \sqrt{\frac{\sum X^2P(X)}{\sum P(X)} - \left[\frac{\sum XP(X)}{\sum P(X)}\right]^2}$$

$$= \sqrt{\sum X^2P(X) - [\sum XP(X)]^2}$$

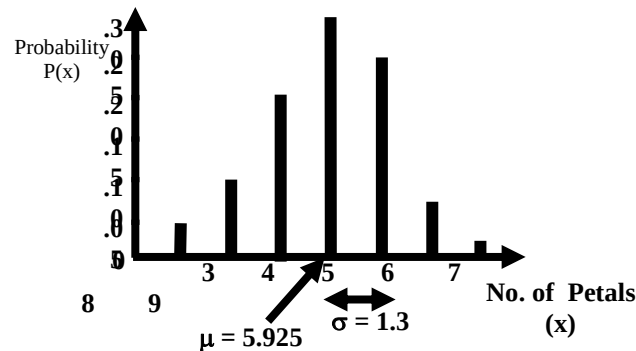
$\left(\because \sum P(X) = 1 \right)$

In the above example

No. of Petals x	P(x)	xP(x)	$x^2P(x)$
$x_1 = 3$	0.05	0.15	0.45
$x_2 = 4$	0.10	0.40	1.60
$x_3 = 5$	0.20	1.00	5.00
$x_4 = 6$	0.30	1.80	10.80
$x_5 = 7$	0.25	1.75	12.25
$x_6 = 8$	0.075	0.60	4.80
$x_7 = 9$	0.025	0.225	2.025
Total	1	5.925	36.925

Hence

$$\begin{aligned}
 \text{S.D.}(X) &= \sqrt{36.925 - (5.925)^2} \\
 &= \sqrt{36.925 - 35.106} \\
 &= \sqrt{1.819} = 1.3
 \end{aligned}$$

Graphical Representation:

Now that we have both the mean and the standard deviation, we are in a position to compute the **coefficient of variation** of this distribution:

Coefficient of Variation

$$\begin{aligned}
 \text{C.V.} &= \frac{\sigma}{\mu} \times 100 \\
 &= \frac{1.3}{5.925} \times 100 \\
 &= 21.9 \%
 \end{aligned}$$

Let us consider another example to understand the concept of discrete probability distribution.

EXAMPLE 1

- Find the probability distribution of the **sum of the dots when two fair dice are thrown**
- Use the probability distribution to find the **probabilities of obtaining (i) a sum that is greater than 8, and (ii) a sum that is greater than 5 but less than or equal to 10.**

SOLUTION

- The sample space S is represented by the following 36 outcomes:

$S = \{(1, 1), (1, 2), (1, 3), (1, 5), (1, 6);$
 $(2, 1), (2, 2), (2, 3), (2, 5), (2, 6);$
 $(3, 1), (3, 2), (3, 3), (3, 5), (3, 6);$
 $(4, 1), (4, 2), (4, 3), (4, 5), (4, 6);$
 $(5, 1), (5, 2), (5, 3), (5, 5), (5, 6);$
 $(6, 1), (6, 2), (6, 3), (6, 5), (6, 6)\}$

Since each of the 36 outcomes is equally likely to occur, therefore each outcome has probability $1/36$.

Let X be the random variable representing the sum of dots which appear on the dice. Then the values of the r.v. are 2, 3, 4, ..., 12.

The probabilities of these values are computed as below:

$f(2) = P(X = 2) = P[\{1, 1\}] = \frac{1}{36}$, as there is only one outcome resulting in a sum of 2,

$$f(3) = P(X = 3) = P[\{(1, 2), (2, 1)\}] = \frac{2}{36},$$

$$f(4) = P(X = 4) = P[\{(1, 3), (2, 2), (3, 1)\}] = \frac{3}{36},$$

Similarly

$$f(5) = \frac{4}{36}, f(6) = \frac{5}{36}, f(7) = \frac{6}{36}, f(8) = \frac{5}{36}, f(9) = \frac{4}{36},$$

$$f(10) = \frac{3}{36}, f(11) = \frac{2}{36} \text{ and } f(12) = \frac{1}{36}.$$

Therefore the desired probability distribution of the r.v X is

x_i	2	3	4	5	6	7	8	9	10	11	12
$f(x_i)$	$\frac{1}{36}$	$\frac{2}{36}$	$\frac{3}{36}$	$\frac{4}{36}$	$\frac{5}{36}$	$\frac{6}{36}$	$\frac{5}{36}$	$\frac{4}{36}$	$\frac{3}{36}$	$\frac{2}{36}$	$\frac{1}{36}$

The probabilities in the above table clearly indicate that if we draw the line chart of this distribution, we will obtain a triangular-shaped graph. The students are encouraged to draw the graph of this probability distribution, in order to be able to develop a visual picture in their minds.

b) Using the probability distribution, we get the required probabilities as follows:

i) $P(\text{a sum that is greater than } 8)$

$$= P(X > 8)$$

$$= P(X=9) + P(X=10) + P(X=11) + P(X=12)$$

$$= f(9) + f(10) + f(11) + f(12)$$

$$= \frac{4}{36} + \frac{3}{36} + \frac{2}{36} + \frac{1}{36} = \frac{10}{36}$$

ii) $P(\text{a sum that is greater than } 5$

but less than or equal to 10)

$$= P(5 < X \leq 10)$$

$$= P(X = 6) + P(X = 7) + P(X = 8)$$

$$+ P(X = 9) + P(X = 10)$$

$$= f(6) + f(7) + f(8) + f(9) + f(10)$$

$$= \frac{5}{36} + \frac{6}{36} + \frac{5}{36} + \frac{4}{36} + \frac{3}{36} = \frac{23}{36}$$

Next, we consider the concept of the DISTRIBUTION FUNCTION of a discrete random variable:

DISTRIBUTION FUNCTION

The distribution function of a random variable X , denoted by $F(x)$, is defined by $F(x) = P(X \leq x)$.

The function $F(x)$ gives the probability of the event that X takes a value LESS THAN OR EQUAL TO a specified value x . The distribution function is abbreviated to *d.f.* and is also called the *cumulative distribution function (cdf)* as it is the cumulative probability function of the random variable X from the smallest value upto a specific value x .

Let us illustrate this concept with the help of the same example that we have been considering --- that of the probability distribution of the sum of the dots when two fair dice are thrown. As explained earlier, the probability distribution of this example is:

x_i	2	3	4	5	6	7	8	9	10	11	12
$f(x_i)$	$\frac{1}{36}$	$\frac{2}{36}$	$\frac{3}{36}$	$\frac{4}{36}$	$\frac{5}{36}$	$\frac{6}{36}$	$\frac{5}{36}$	$\frac{4}{36}$	$\frac{3}{36}$	$\frac{2}{36}$	$\frac{1}{36}$

The term 'distribution function' implies the cumulating of the probabilities similar to the cumulation of frequencies in the case of the frequency distribution of a discrete variable.

x_i	2	3	4	5	6	7	8	9	10	11	12
$f(x_i)$	$\frac{1}{36}$	$\frac{2}{36}$	$\frac{3}{36}$	$\frac{4}{36}$	$\frac{5}{36}$	$\frac{6}{36}$	$\frac{5}{36}$	$\frac{4}{36}$	$\frac{3}{36}$	$\frac{2}{36}$	$\frac{1}{36}$
$F(x_i)$	$\frac{1}{36}$	$\frac{3}{36}$	$\frac{6}{36}$	$\frac{10}{36}$	$\frac{15}{36}$	$\frac{21}{36}$	$\frac{26}{36}$	$\frac{30}{36}$	$\frac{33}{36}$	$\frac{35}{36}$	$\frac{36}{36}$

If we are interested in finding the probability that we obtain a sum of five or less, the column of cumulative probabilities immediately indicates that this probability is $\frac{10}{36}$.